Predicting Forest Wildfire Risk

by Aryan Singh

Wildlife in Forest

1 **Unique Patterns**

Large-scale fires occur annually in boreal forests and remote wilderness.

2 **Contributing Factors**

Vegetation, weather, and ignition sources drive wildfire risk.

Usinq satellite imagery and machine learning to predict wildfire risk in Forest.Accurate predictions aid in disaster mitigation and efficient resource allocation.

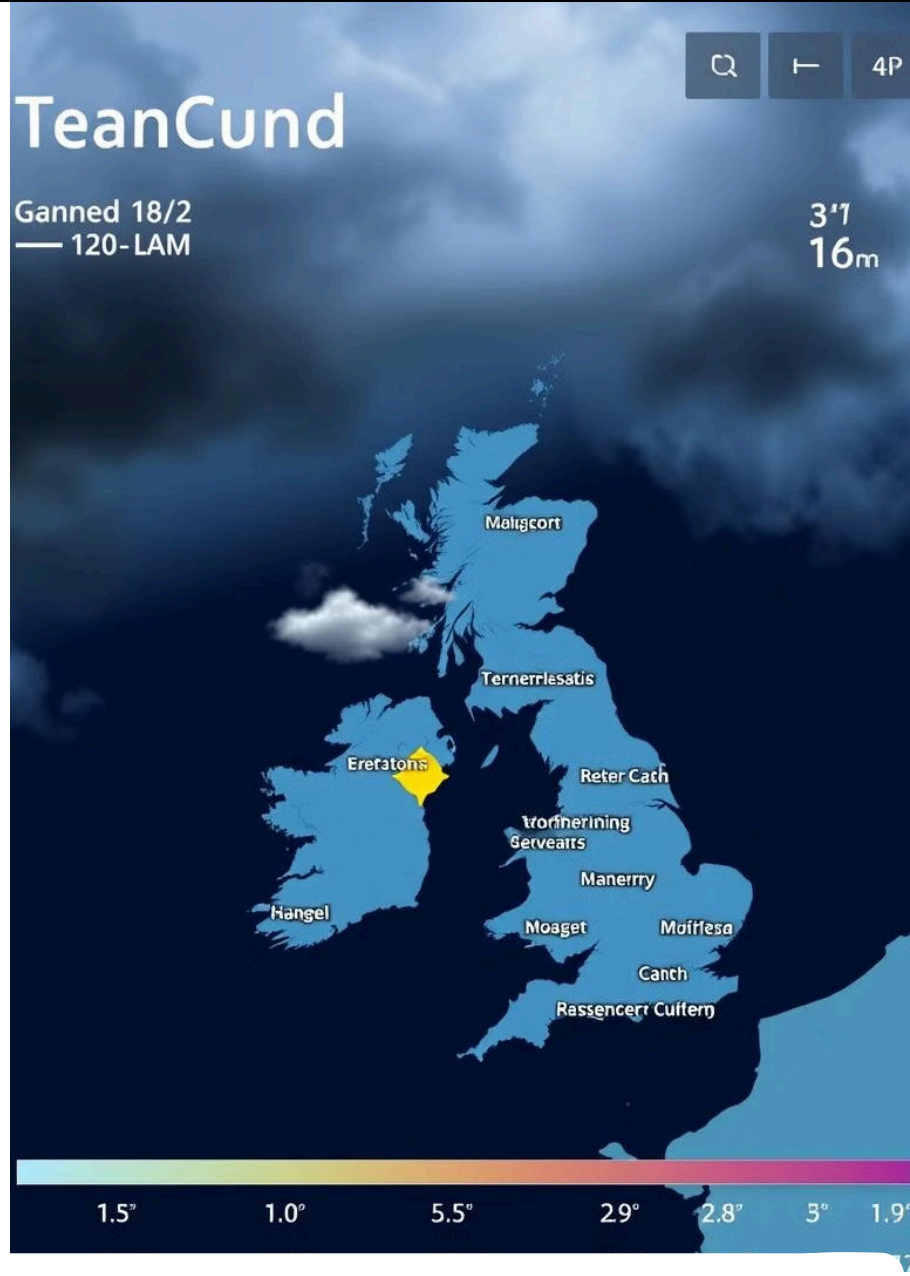
Traditional Prediction Limitations

Delayed Data

Reliance on weather data and historical records can be delayed.

Inaccurate Observations

Human observations are inaccurate, especially in remote areas.



Advantages of Satellite Imagery

Real-time Insights

High-resolution data on vegetation, thermal anomalies, and soil moisture.

Cloud Penetration

SAR enables fire detection even through cloud cover.

Data Sources

Satellite Data

- Landsat 8 & 9: Vegetation and burn severity.
- Sentinel-2: High-resolution fire risk classification.
- MODIS & VIIRS: Fire perimeters and monitoring.
- Sentinel-1 & ALOS-2 (SAR): Vegetation moisture.

Ground Data

- ERA5: Historical and real-time weather.
- NOAA NWS: Near real-time weather data.
- Forest Fire Service: Ignition source data.

Satellite Data Collection & Processing

1

Acquisition

Satellite images are acquired.

2

Storage

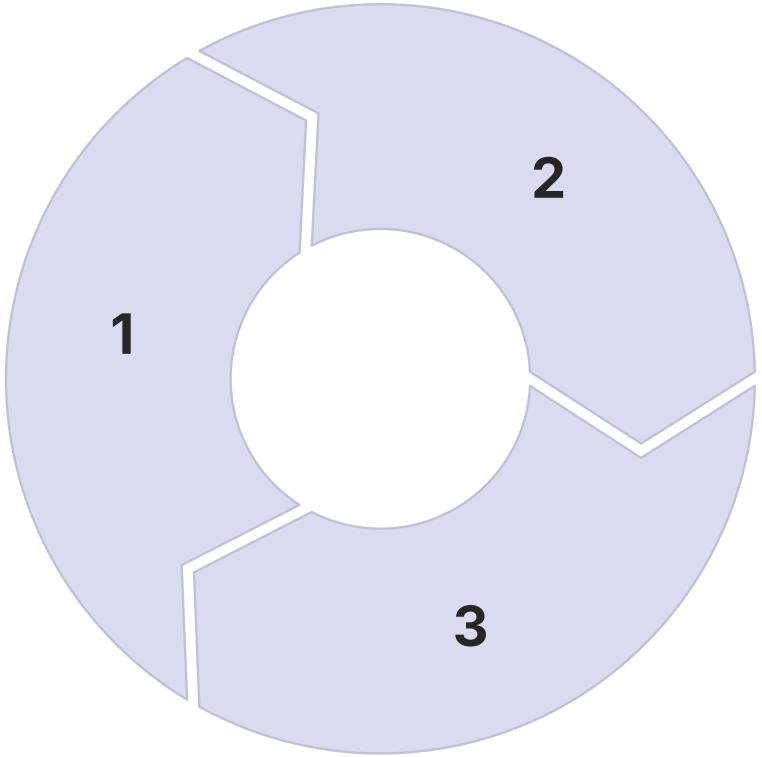
Data is stored.

3

Processing

Data is processed.

Methodology



Preprocessing

Band selection and cloud removal.

Model Development

Hybrid deep learning model (CNN-LSTM).

Visualization

Web-based GIS dashboard.

Expected Outcomes

1

Fire Risk Classification

High, Moderate, or No Risk.

2

Performance Metrics

Model accuracy and reliability.

3

Operational Tool

Actionable insights.

Conclusion

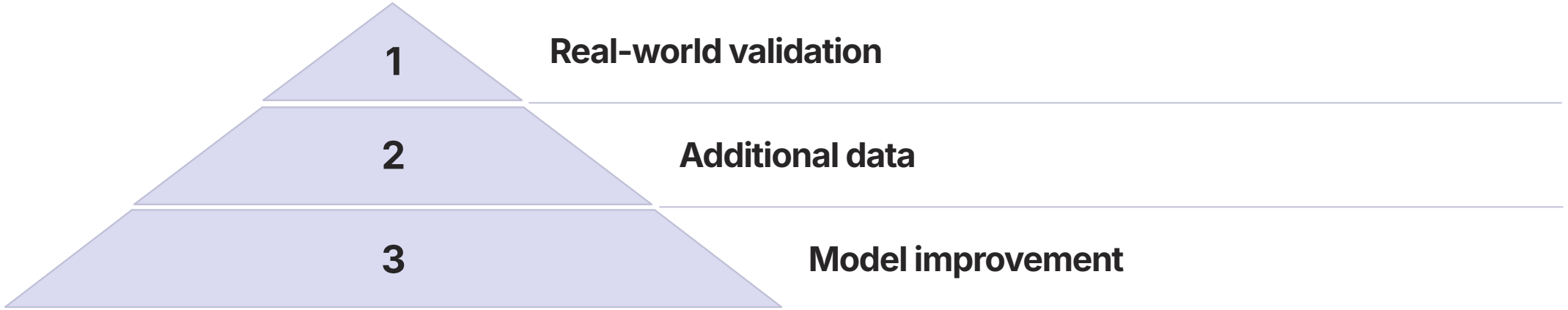
90%

Enhancing wildfire prediction capabilities.

50%

Continuous model improvement.

Future Work


1
Real-world validation
2
Additional data
3
Model improvement